RAAN: an open, reproducible laboratory for orbital data — SGP4 digital twin, collision-risk forecasting on ESA conjunction data, and manoeuvre detection from public element sets

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Independent student project, Madrid · 12 November 2026 · code and data pipeline: https://github.com/didimacveg/kessler · Research project, not for operational use.
Written and coded with substantial help from an AI coding assistant (Claude Code); the author is responsible for every claim.

Abstract. This report documents three experiments built on public orbital data, each with its baseline reproduced before any model is trained. (i) The browser propagator reproduces the 666 reference state vectors of the SGP4 verification set to 1.2×10^{-7} km, and to 3.5×10^{-5} km when initialised from OMM fields as the application does; it renders the 20,152 objects for which bulk elements are public. (ii) On ESA's Collision Avoidance Challenge dataset, the naive "latest risk" forecast is reproduced exactly (L = 0.694). A gradient-boosted gate model reaches L = 0.627 on the test set, but cross-validation on the training set ranks it below the naive forecast (1.23 vs 0.80) and the bootstrap interval of the difference contains zero; a direct regressor and a recurrent model are clearly worse. No model is claimed to improve on the baseline, and the role of the documented selection bias of the dataset is quantified. (iii) On 803 operator-reported manoeuvre events of 15 satellites, evaluated leave-one-satellite-out, a gradient-boosted classifier reaches F1 = 0.81, a physics baseline 0.74 and CUSUM 0.52; a detector using only the step in semi-major axis reaches 0.79, and the same detections placed at random times 0.09. All figures are regenerated from source by one command.

1 Problem

More than 35,194 catalogued objects orbit the Earth. Operators receive conjunction data messages (CDMs) days before a close approach and typically have to commit to an avoidance manoeuvre about two days ahead, while the estimated risk is still changing. Separately, anyone outside an operator sees manoeuvres only through their effect on published orbital elements. Both are forecasting and detection problems on noisy, irregular series, and both have public data. This project asks two narrow questions — can a learned model beat the naive risk forecast, and how well can reported manoeuvres be recovered from public element sets — and is deliberately strict about what counts as an answer.

It is not an operational system. Operational collision avoidance relies on tracking data, covariance realism and continuous processes provided by organisations such as GMV (Focusoc), Neuraspace, LeoLabs or Slingshot; public viewers such as KeepTrack are more complete than the visualisation built here; and the open-source *kessler* library addresses probabilistic CDM forecasting. The contribution is a small, fully reproducible set of baselines and negative results.

2 Data

2.1 Orbital elements

General-perturbations (GP) mean elements are taken from CelesTrak in OMM CSV form together with the SATCAT. The TLE format is not used: new objects have received 6-digit catalogue numbers since July 2026 and cannot be represented in it; 1,373 such objects are in the snapshot. CelesTrak has no "all objects" query, so the snapshot is the union of the active list, analyst objects and three debris clouds: 20,152 objects, against 35,194 on orbit in the SATCAT. The median element set is 16 h old at download. CelesTrak's usage policy (one download per two-hour update, no retries on HTTP errors) is implemented in the fetcher.

2.2 Conjunction data

The ESA Collision Avoidance Challenge dataset (Zenodo, DOI 10.5281/zenodo.4463683, CC BY 4.0) holds 187,118 anonymised CDMs: 13,154 training and 2,167 test events. The quantity to predict is the log10 collision probability in the last CDM of each event. Test inputs stop two days before the time of closest approach (TCA); the target CDM is within one day of it. ESA states that both sets were hand-picked to over-represent high-risk events. The consequence is

measurable: among training events filtered like the test set, 66 of 8,293 end at or above 10^{-6} (0.8%); in the test set 150 of 2,167 do (6.9%).

2.3 Manoeuvre labels

The benchmark of Shorten et al. provides TLE histories (38,622 element sets, about one per day) and 941 reported manoeuvre times for 15 satellites: ten LEO altimetry and geodesy missions whose operators publish manoeuvres through the International DORIS Service, and five Fengyun geostationary satellites. The repository has no licence file; it is cloned at a pinned commit at build time and only derived series are published.

3 Methods

3.1 Propagation and its verification

Positions are SGP4 propagations of the published elements in the TEME frame, computed in a Web Worker (satellite.js) for the visualisation and with the *sgp4* Python package in the pipeline. Both are tested against the verification file of Vallado et al. (33 element sets, 666 state vectors): the maximum position difference is 1.2×10^{-7} km for satellite.js and 1.2×10^{-7} km for the Python package. One deliberately invalid element set in that file must be rejected, and is. The application does not read TLEs but OMM fields; through that route the largest difference is 3.5×10^{-5} km, on a 14-day orbit of eccentricity 0.97, because the library reads the epoch to the millisecond. For the same reason the absolute time of a displayed position is good to about a millisecond, a few metres along-track. The display interpolates between worker key-frames on the GPU (direction and radius separately, key-frames at most 240 s apart). Nothing shown is an observation.

3.2 Final-risk prediction

The official score is

$$L = MSE_{HR} / F_2$$

where an event is high-risk if its final risk $r \geq -6$, F_2 is computed on that label over all events, MSE_{HR} is the mean squared error over truly high-risk events, and predictions below -6 are clipped to -6.001. Two baselines from the challenge paper are reproduced: constant risk (-5) and the latest-risk prediction (LRP), the clipped last risk known two days before TCA.

Three learned models are compared. *Gate + value*: a LightGBM classifier estimates the probability that the event ends high-risk from 15+ event features (latest CDM, covariance sizes, observation statistics, and summaries of the risk history); events above a threshold τ receive a value from a two-parameter linear function of the latest risk fitted on truly high-risk training events; all others receive -6.001. *Direct regression*: LightGBM on the clipped target. *Sequence model*: a GRU (32 hidden units) over the last eight CDMs with a gate head and a value head, three seeds averaged. Training uses events whose final CDM is within a day of TCA; each is also presented as it looked three and four days out. τ and all model selection use 5-fold cross-validation grouped by event, on the training set only; test labels are never used to fit or tune anything. The test set was, however, scored more than once during development (for instance before and after a bug fix in one feature), so it is not a pristine hold-out. Two cross-validated losses are given for the gate: one with τ tuned on the very predictions it scores, which flatters the gate, and a nested one in which τ and the value model are chosen inside each fold. Uncertainty is a 2,000-resample event bootstrap, paired against LRP.

3.3 Manoeuvre detection

For consecutive element sets the pipeline computes the change in mean semi-major axis a and inclination i, and the RTN residual: the earlier set propagated to the epoch of the later one, minus the later set's own position, in radial, along-track and cross-track components. Mean elements are compared directly, as recommended by Shorten et al.

Physics baseline. A level-shift statistic: the median of the three sets after an interval minus the median of the three before, minus the secular drift estimated from the previous 30 intervals, expressed in robust (MAD) sigmas of the satellite's own recent history. A single bad element set produces a spike and cancels; a burn produces a step. The score is the larger of the a and i statistics. *CUSUM.* Page's two-sided cumulative sum on standardised day-to-day changes of a and i. *Classifier.* LightGBM on the statistics above, RTN residuals, raw magnitudes and their values two sets before and after.

Evaluation is at event level. Reported burns closer than 1.5 days form one event. 54 reported burns fall before the first or after the last element set of their satellite's history; nothing can detect them and they are not scored. Consecutive flagged intervals form one detection, which explains an event if the burn lies between two days before its start and its end; matching is one-to-one. Because that window is days wide, the floor is measured rather than assumed: each detector's own detections, moved to random times, are scored with the same rule. Thresholds, and the classifier itself, are chosen leave-one-satellite-out, so the figures for a satellite never use its labels.

3.4 Reporting

A periodic briefing is assembled by small agents that read the artefacts above and register facts (value, source file). A reviewer rejects any statement citing a missing fact or containing a number not present in its cited facts; an optional language model may rephrase a paragraph only if the same check passes. Agents do not compute.

4 Results

4.1 Final-risk prediction

The LRP baseline is reproduced exactly: L = 0.694, $\mathrm{MSE}_{\mathrm{HR}}$ = 0.513, F_2 = 0.739 (published: 0.694, 0.513, 0.739). Table 1 lists all models.

Table 1. Test-set scores (2,167 events, 150 high-risk). CV = cross-validated L on training events. FN = missed high-risk events, FP = false alarms. Last column: share of bootstrap resamples in which the model beats LRP.

Model	L	95% int.	CV	$\mathrm{MSE}_{\mathrm{HR}}$	F_2	FN	FP	P(<LRP)
Constant risk (−5)	2.504	1.86–3.34	—	0.679	0.271	0	2017	0%
Latest risk (naive forecast)	0.694	0.46–0.96	0.80	0.513	0.739	35	63	—
LightGBM gate + value	0.627	0.44–0.86	0.98	0.466	0.743	13	185	86%
LightGBM direct regression	2.374	1.79–3.06	6.30	1.408	0.593	23	344	0%
GRU sequence model	1.318	0.90–1.83	1.57	0.884	0.670	35	143	0%
sesc (published)	0.556	—	—	0.407	0.733	—	—	—
dietmarw (published)	0.571	—	—	0.437	0.765	—	—	—
Magpies (published)	0.585	—	—	0.441	0.753	—	—	—

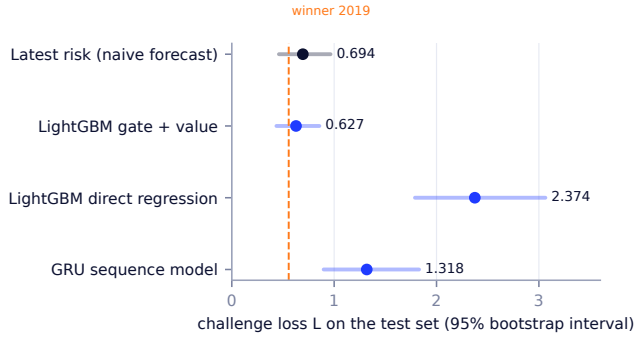


Figure 1. Challenge loss with bootstrap intervals. Dashed: winning entry of the 2019 competition.

The gate model has the lowest test loss of the models trained here (0.627). It is nevertheless not reported as an improvement, for two reasons. First, the selection rule — lowest cross-validated loss on training data — prefers LRP: 0.80 against 0.98 for the gate, and 1.23 once the threshold and the value model are also chosen inside each fold (the CV column of Table 1 shows the first, more favourable figure). Second, the 95% interval of the paired difference is [-0.19, 0.05]; the gate wins in 86% of resamples. Direct regression (L = 2.37) is pulled to the low-risk floor by the 99% of low-risk training events. The GRU scores 1.32: it misses 35 high-risk events (naive forecast: 35) and raises 143 false alarms against 63.

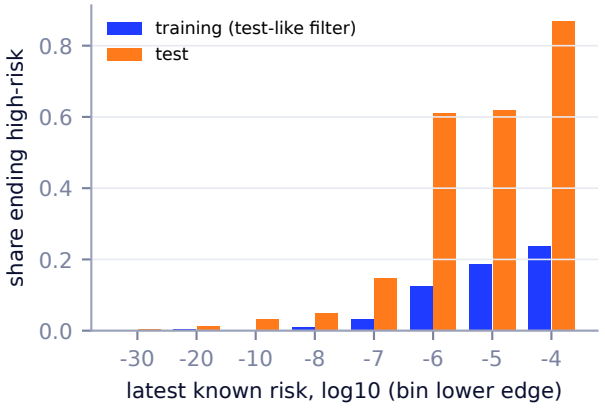


Figure 2. Probability that an event ends high-risk given its latest known risk. The same input has a different outcome distribution in the two sets.

Why test and cross-validation disagree. Figure 2 shows the selection bias directly. An event whose latest risk lies in [−6, −5) ends

high-risk in 13% of training cases and 61% of test cases. A classifier trained on the former is therefore under-confident on the latter (expected calibration error 0.037 on test against 0.005 out-of-fold), and a threshold tuned for 0.8% prevalence produces many false alarms in cross-validation but is cheap when 6.9% of events are positive. The apparent test gain is thus largely a property of how the split was constructed.

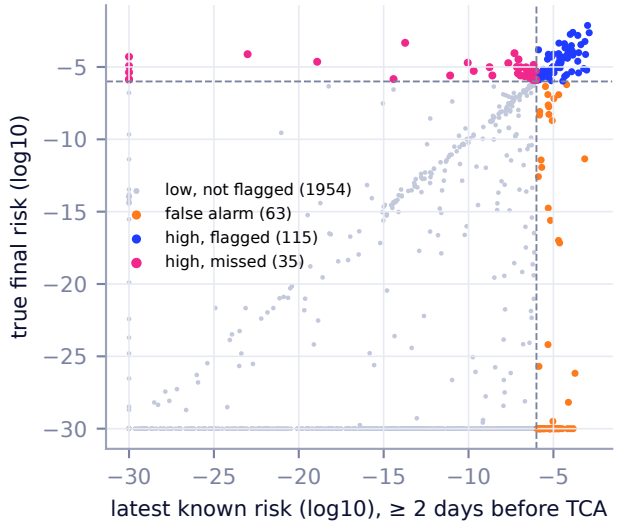


Figure 3. Test events under the naive forecast. Events at the left edge that end above the dashed line carried no risk signal two days out.

Error analysis. LRP misses 35 of 150 high-risk events. 16 had a latest risk between −7 and −6 and are plausibly learnable; 11 were below −10, of which 5 sat at the dataset floor of −30 together with 1,302 other test events. For those, the risk appeared after new tracking data arrived inside the last two days; no input available at decision time distinguishes them.

4.2 Manoeuvre detection

Table 2. Event-level precision (P), recall (R) and F1, leave-one-satellite-out, 803 events.

Detector	P	R	F1	P	R	F1	P	R	F1
		LEO (10 sat.)			GEO (5 sat.)				all
Physics (a + i step)	0.76	0.64	0.69	0.88	0.90	0.89	0.79	0.70	0.74
CUSUM	0.57	0.44	0.50	0.40	0.86	0.54	0.49	0.54	0.52
GB classifier	0.83	0.75	0.79	0.81	0.93	0.87	0.83	0.80	0.81

The classifier is best overall (F1 = 0.81); in GEO the physics baseline scores 0.89 against 0.87. CUSUM is worst: it integrates every noisy stretch of element sets into an alarm. For scale, the same detections at random times score F1 = 0.09 (classifier), 0.12 (physics) and 0.23 (CUSUM, which fires most often).

Table 3. Ablation of the physics baseline (all satellites).

Signal	P	R	F1
Single-interval jumps + RTN residual	0.24	0.65	0.35
Step in semi-major axis	0.85	0.74	0.79
Step in a or i (baseline)	0.79	0.70	0.74
Step in a or i, RTN-confirmed	0.70	0.63	0.66

The ablation (Table 3) carries the main lesson. Thresholding single-interval jumps and RTN residuals — the obvious first design — yields F1 = 0.35, because every bad element set triggers twice. The level-shift statistic on semi-major axis alone reaches 0.79; adding inclination lowers it to 0.74, and requiring RTN confirmation to 0.66. The classifier therefore adds about 2 F1 points over the best one-signal rule, not 7.

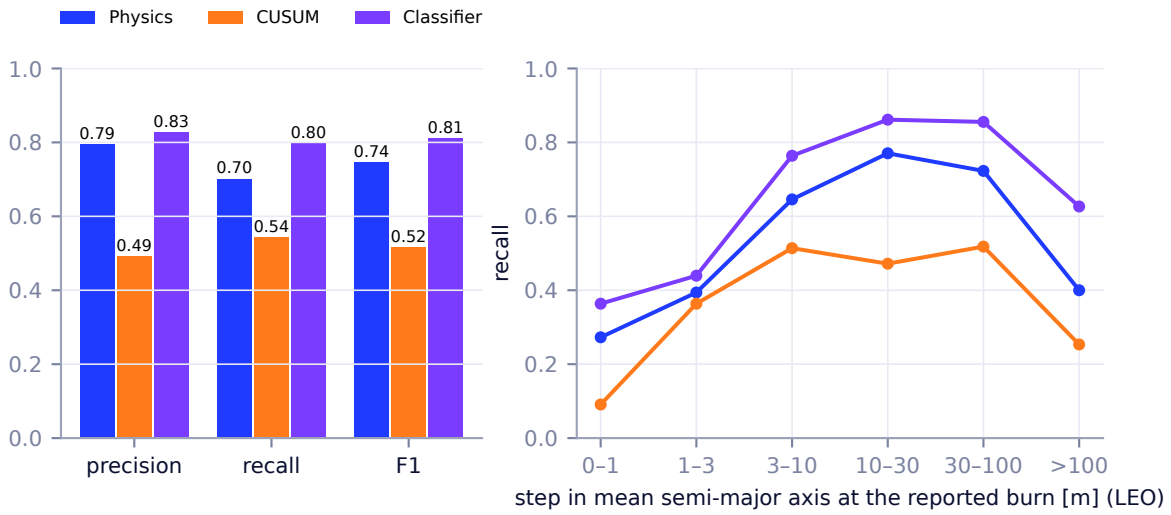


Figure 4. Left: event-level scores over all satellites. Right: recall against the size of the step in mean semi-major axis at the reported burn (LEO).

Detectability is set by burn size (Figure 4, right). The step statistic itself has a noise floor: on 23,473 LEO intervals more than 6 days from any reported burn its absolute value has a median of 0.4 m and a 95th percentile of 4.1 m. Accordingly the classifier finds 29 of 66 burns that moved the mean semi-major axis by 1 to 3 m, and 4 of the 11 below 1 m, too few to say more than that such burns are at the level of the noise. Recall also drops above 100 m, for a different reason: 24 of the 28 large burns missed by the classifier have another reported burn within four days. These are orbit-change campaigns; at one element set per day they merge into one detection, which one-to-one matching credits once. Changing the matching lag from two days to one or three changes the classifier's F1 by 0.017.

Table 4. F1 per satellite (held out in turn). n = manoeuvre events.

Satellite	n	Physics	CUSUM	Classifier
CryoSat-2 (LEO)	150	0.85	0.58	0.91
Fengyun-2D (GEO)	22	0.88	0.41	0.87
Fengyun-2E (GEO)	48	0.92	0.46	0.86
Fengyun-2F (GEO)	66	0.93	0.63	0.94
Fengyun-2H (GEO)	12	0.76	0.50	0.71
Fengyun-4A (GEO)	45	0.84	0.71	0.82
Haiyang-2A (LEO)	56	0.50	0.40	0.58
Jason-1 (LEO)	90	0.61	0.54	0.69
Jason-2 (LEO)	69	0.81	0.69	0.88
Jason-3 (LEO)	38	0.66	0.49	0.74
SARAL (LEO)	54	0.56	0.48	0.83
Sentinel-3A (LEO)	59	0.85	0.33	0.96
Sentinel-3B (LEO)	46	0.88	0.60	0.91
Sentinel-6A (LEO)	11	0.57	0.74	0.64
TOPEX (LEO)	37	0.27	0.18	0.46

Per-satellite results (Table 4) vary more than the method averages: from 0.46 to 0.96 for the classifier. The weakest case is TOPEX; why — tracking quality of its era, burn sizes, or both — was not investigated.

5 Discussion

The two experiments point the same way. In both, most of the achievable performance comes from one physically meaningful quantity — the latest risk, the step in semi-major axis — and the learned models mostly re-weight it. The classifier's most important input is the semi-major-axis step (36% of split gain); the gate's is the

mean of the risk history (27%). Where the information is absent from the inputs — risk that appears in the last two days, burns below the noise of the element sets — no model recovers it.

The second lesson is about evaluation rather than modelling. Had the test score alone been reported, the collision experiment would read as a 10% improvement over the baseline that only twelve of ninety-seven teams beat. The cross-validation estimate, the bootstrap interval and the transition analysis each tell a more careful story, and they cost a few lines of code. The same holds for the manoeuvre ablation: without it, the classifier would appear to add 7 F1 points instead of 2.

6 Limitations

Collision. With 150 high-risk test events every interval is wide. The data are from 2015–2019, anonymised, from one agency's missions, and "risk" is a computed probability: no collision is in the data. Test labels were never used for fitting or tuning, but the test set was scored several times during development and the challenge paper, which reports what worked, was read beforehand. The bootstrap resamples test events only; it does not include the variance of training. Nothing here transfers to operational prevalence.

Manoeuvres. Fifteen well-tracked satellites performing impulsive station-keeping. Low-thrust electric propulsion, which dominates today's large constellations, is not covered. Labels are operator reports: an unreported burn counts as a false alarm. Thresholds and model fits are out-of-sample, but the detector design (window lengths, the level-shift statistic, the matching rule) was iterated on these same satellites. The classifier uses the two element sets after an interval and the step statistic three, so it is a retrospective detector. The level-shift statistic needs three later element sets, so a detection is available about three days after the burn.

Visualisation. 20,152 of 35,194 on-orbit objects are shown; most debris and rocket bodies require a Space-Track account. SGP4 accuracy degrades by kilometres per day away from the element epoch. Frame rate was measured only under software rendering; on slow devices the page lowers its resolution and says so.

7 Future work

(1) A forward archive of daily CelesTrak snapshots started with this project; after about a month it allows the manoeuvre detectors to run on any object, initially unlabelled. (2) Supplemental GP data fitted to operator ephemerides (e.g. Starlink) would allow a study of low-thrust orbit raising. (3) For conjunctions, predicting a calibrated distribution of the final risk rather than a point value, and evaluating with proper scoring rules under an explicit prevalence assumption, would separate model quality from the split's bias. (4) Covariance-aware features (for instance the evolution of miss distance in units of combined position uncertainty) were not explored.

Reproducibility

`make setup data reproduce` rebuilds every table and figure from the public sources (about ten minutes, 221 MB download). Collision results: code e7b7978, run 2026-11-12; manoeuvre results: code e7b7978, benchmark commit 5ea648955c; catalogue snapshot 2026-11-11T22:42:12Z. Seeds are fixed; tests cover the metric definitions, the protocol filters, the propagators and the reporting checks.

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Orbital data: CelesTrak, derived from 18 SDS / Space-Track.org. Conjunction data: European Space Agency, CC BY 4.0. Manoeuvre labels: International DORIS Service via Shorten et al. Research project, not for operational use.